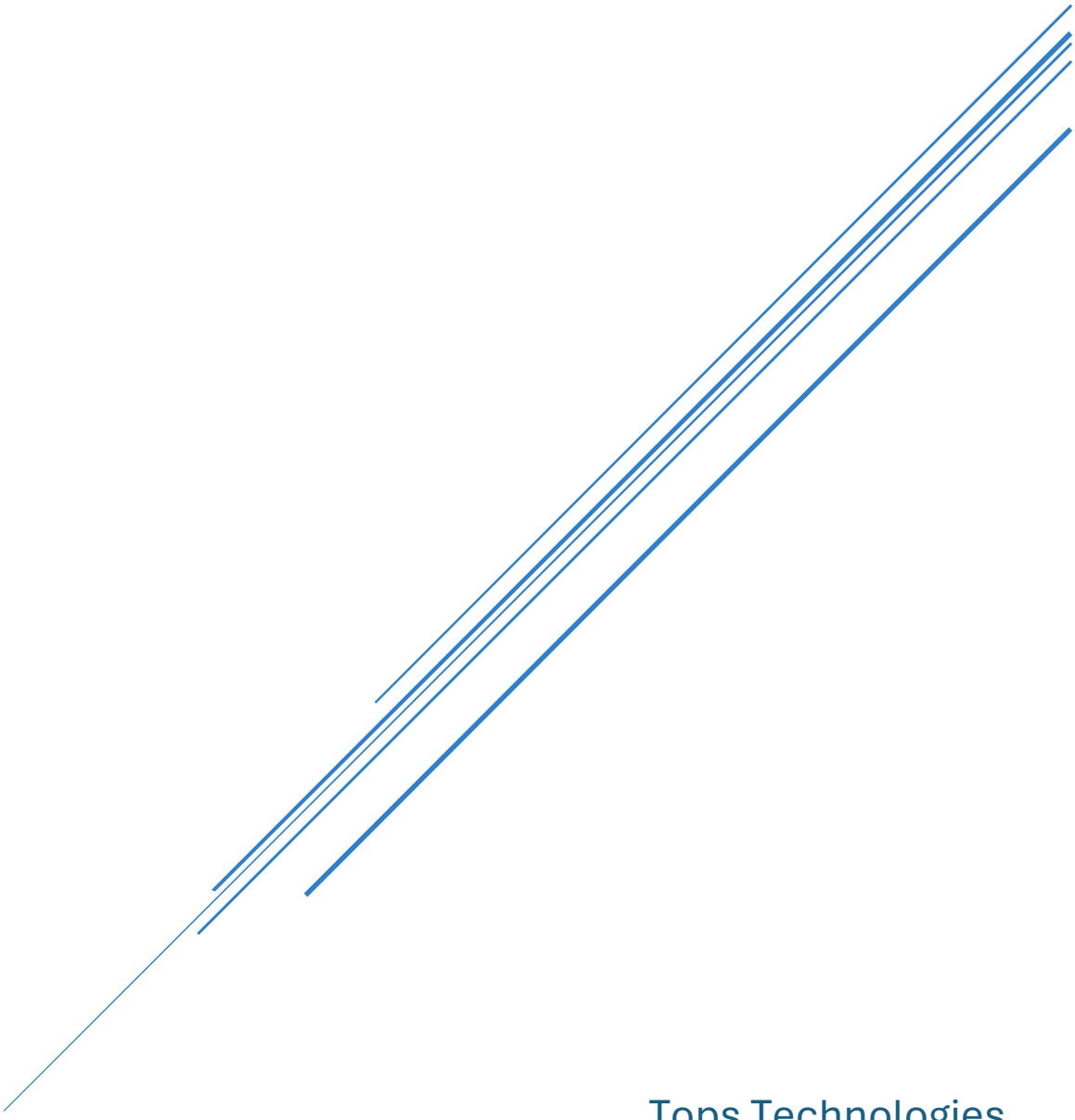
DAY – 3 INTRODUCTION TO IP ADDRESSING

Assignment - 14



Tops Technologies
Cyber Security

1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

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IPv4	IPv6
Uses a 32-bit address.	Uses a 128-bit address.
Written in decimal format, e.g., 192.168.1.1.	Written in hexadecimal format, e.g., 2001:db8::1.
Provides approximately 4.3 billion addresses.	Provides a very large number of addresses.
Uses four numbers separated by dots (.).	Uses eight groups separated by colons (:).
Supports broadcast communication.	Does not support broadcast; it uses multicast instead.
NAT is commonly used to conserve IP addresses.	NAT is generally not required because of its huge address space.

2. **Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.**

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Device	Static IPv4 Address	Subnet Mask	Default Gateway
Laptop	192.168.10.2	255.255.255.0	192.168.10.1
Phone	192.168.10.3	255.255.255.0	192.168.10.1
Smart TV	192.168.10.4	255.255.255.0	192.168.10.1
Printer	192.168.10.5	255.255.255.0	192.168.10.1

Explanation:

- **IP Address: A unique address assigned to each device.**
- **Subnet Mask: 255.255.255.0 identifies the network portion of the IP address.**
- **Default Gateway: 192.168.10.1 is the home Wi-Fi router's IP address, which connects devices to other networks, such as the internet.**

3. **Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.**

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1. Given Information

- **Network: 192.168.1.0**
- **CIDR subnet mask: /27**
- **Subnet mask: 255.255.255.224**
- **Default Class C mask: /24**

2. Calculate the Number of Subnets and Hosts

Step 1: Number of borrowed bits

\[27-24=3\text{ bits} \]

Number of subnets:

\[2^3=\boxed{8\text{ subnets}} \]

Step 2: Number of host bits

\[32-27=5\text{ bits} \]

Total addresses per subnet:

\[2^5=32 \]

Usable hosts per subnet:

\[32-2=\boxed{30\text{ hosts}} \]

We subtract 2 because the network address and broadcast address cannot be assigned to devices.

3. First Three Valid Subnet Ranges

The block size is \(256-224=32\), so each subnet starts 32 addresses after the previous one.

Details	Subnet 1	Subnet 2	Subnet 3
Network address	192.168.1.0	192.168.1.32	192.168.1.64
First usable host	192.168.1.1	192.168.1.33	192.168.1.65
Last usable host	192.168.1.30	192.168.1.62	192.168.1.94
Broadcast address	192.168.1.31	192.168.1.63	192.168.1.95
Usable hosts	30	30	30

4. Final Answer

- **Total valid subnets: 8**
- **Usable hosts per subnet: 30**
- **Total usable hosts across all subnets: \(8 \times 30 = 240\)**
- **Subnet mask: 255.255.255.224**
- **Block size: 32**

4. **You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.**
   Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

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Given Information

- **IP Address: 10.0.5.17**
- **Subnet Mask: 255.255.255.0**
- **CIDR Notation: /24**

Calculations

1. Network Address

Perform a bitwise AND between the IP address and subnet mask.

10.0.5.17 AND 255.255.255.0 = 10.0.5.0

Network Address: 10.0.5.0

2. Broadcast Address

Since the subnet mask is /24, the last 8 bits are reserved for host addresses. Set all host bits to 1.

Broadcast Address: 10.0.5.255

3. Valid Host Address Range

- **First usable IP: 10.0.5.1**
- **Last usable IP: 10.0.5.254**

Final Answer

Item	Result
IP Address	10.0.5.17
Subnet Mask	255.255.255.0
CIDR Notation	/24
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
Valid Host Range	10.0.5.1 – 10.0.5.254
Total Usable Hosts	254

5. **Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.**

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